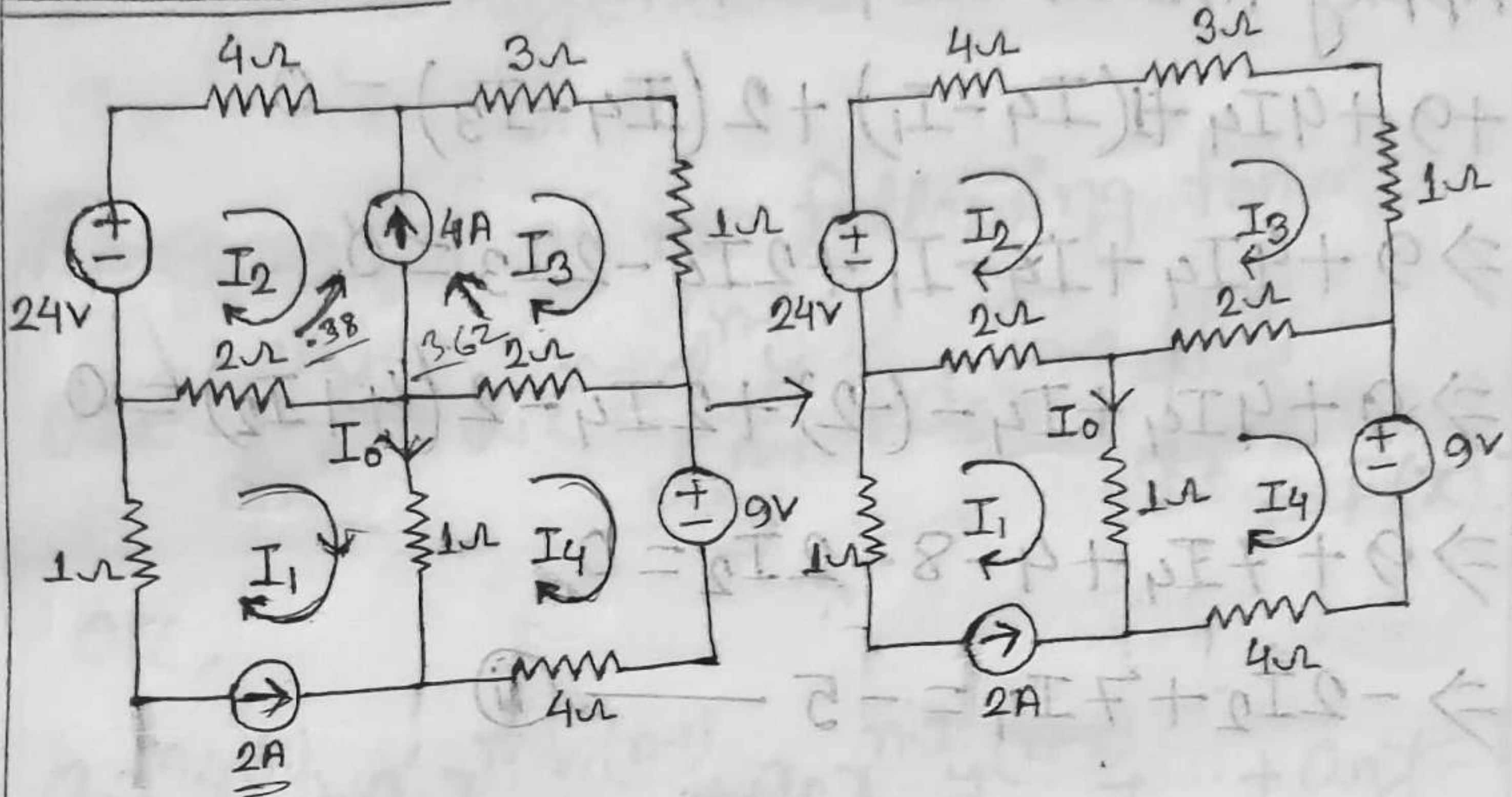


TOPIC :
MESH
ANALYSIS

Prob 3.38:-



$$I_1 = -2A, \quad I_0 = I_2 - I_4 = \frac{0V}{80\Omega} = 0A$$

$$I_3 - I_2 = 4$$

$$\Rightarrow I_3 = 4 + I_2$$

Apply KVL at super mesh.

$$-24 + 4I_2 + 3I_3 + I_3 + 2(I_3 - I_4) + 2(I_2 - I_1) = 0$$

$$\Rightarrow -24 + 4I_2 + 3I_3 + I_3 + 2I_3 - 2I_4 + 2I_2 - 2I_1 = 0$$

$$\Rightarrow -24 + 6I_2 + 6I_3 - 2I_4 - 2I_1 = 0$$

$$\Rightarrow -24 + 6I_2 + 6(4 + I_2) - 2I_4 - 2(-2) = 0$$

$$\Rightarrow -24 + 6I_2 + 24 + 24I_2 - 2I_4 + 4 = 0$$

$$\Rightarrow 30I_2 - 2I_4 = -10 \text{ mA} \quad \text{①}$$

Apply KVL at Loop I_4 ,

$$+9 + 4I_4 + 1(I_4 - I_1) + 2(I_4 - I_3) = 0$$

$$\Rightarrow 9 + 4I_4 + I_4 - I_1 + 2I_4 - 2I_3 = 0$$

$$\Rightarrow 9 + 4I_4 + I_4 - (-2) + 2I_4 - 2(4 + I_2) = 0$$

$$\Rightarrow 9 + 7I_4 + 4 - 8 - 2I_2 = 0$$

$$\Rightarrow -2I_2 + 7I_4 = -5 \quad \text{--- (ii)}$$

$$\therefore I_2 = -\frac{40}{103} = -0.38 \text{ A}$$

$$\therefore I_4 = -\frac{85}{103} = -0.82 \text{ A}$$

$$\therefore I_3 = 4 + I_2$$

$$= 4 + (-0.38) = 3.62 \text{ A}$$

$$I_0 = I_1 - I_4$$

$$= -2 - (-0.82)$$

$$\therefore I_0 = -1.18 \text{ A}$$

(i) Am

Q17, 38:-

$$I_1 = -2A$$

$$I_0 = I_1 - I_4$$

$$I_3 - I_2 = 4$$

$$\Rightarrow -I_2 + I_3 + 0 \cdot I_4 = 4 \quad \text{--- (i)}$$

Apply KVL at super mesh,

$$-24 + 4I_2 + 3I_3 + I_3 + 2(I_3 - I_4) + 2(I_2 - I_1) = 0$$

$$\Rightarrow -24 + 4I_2 + 3I_3 + I_3 + 2I_3 - 2I_4 + 2I_2 - 2I_1 = 0$$

$$\Rightarrow -24 + 6I_2 + 6I_3 - 2I_4 - 2(-2) = 0 \quad \text{--- (ii)}$$

$$\Rightarrow -6I_2 + 6I_3 - 2I_4 = 20 \quad \text{--- (ii)}$$

Apply KVL at mesh I_4 , I

$$+9 + 4I_4 + 1(I_4 - I_1) + 2(I_4 - I_3) = 0$$

$$\Rightarrow 9 + 4I_4 + I_4 - I_1 + 2I_4 - 2I_3 = 0$$

$$\Rightarrow 9 + 7I_4 - (-2) - 2I_3 = 0$$

$$\Rightarrow -2I_3 + 7I_4 = -11$$

$$\Rightarrow 0 \cdot I_2 - 2I_3 + 7I_4 = -11 \quad \text{--- (iii)}$$

(i)

$$\therefore I_2 = -0.425A$$

$$\therefore I_3 = 3.575A$$

$$\therefore I_4 = -0.55A$$

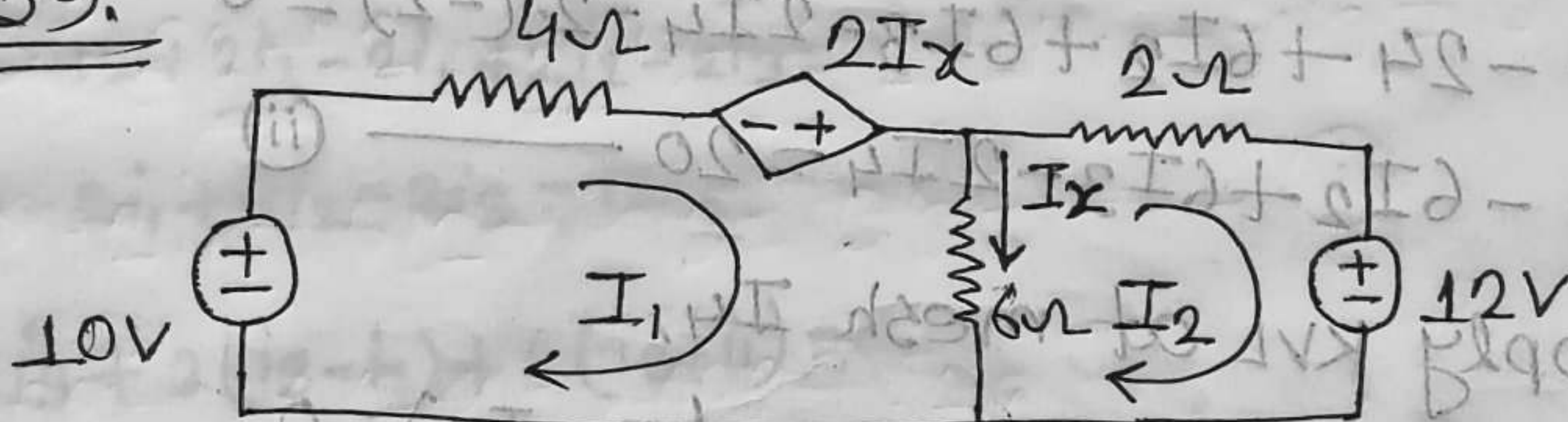
$$\therefore I_0 = I_1 - I_4$$

$$= -2 - (-0.55)$$

$$= -1.45A$$

Ans

39.



$$I_x = I_1 - I_2$$

Apply KVL at Loop I_1 ,

$$-10 + 4I_1 - 2(I_1 - I_2) + 6(I_1 - I_2) = 0$$

$$\Rightarrow -10 + 4I_1 - 2I_1 + 2I_2 + 6I_1 - 6I_2 = 0$$

$$\Rightarrow 8I_1 - 4I_2 = 10 \quad \text{--- (i)}$$

Apply KVL at Loop I_2 ,

$$2I_1 + 12 + 6(I_2 - I_1) = 0$$

$$\Rightarrow 2I_2 + 12 + 6I_2 - 6I_1 = 0$$

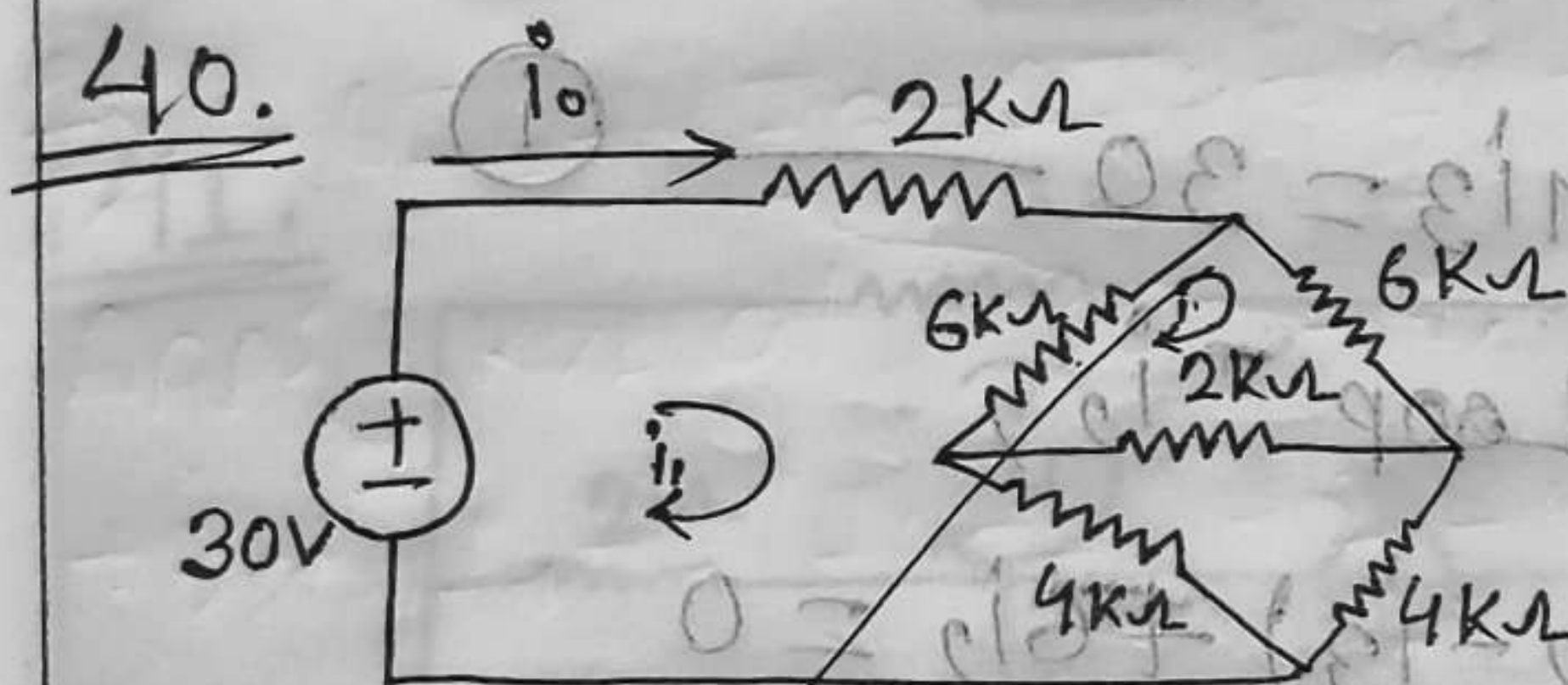
$$\Rightarrow -6I_1 + 8I_2 = -12$$

$$\therefore I_1 = 0.8A$$

$$\therefore I_2 = -0.9A$$

$$0 = (i_1 - i_2) \underline{Am}$$

$$0 = i_1 R - i_2 R + i_1 R - i_2 R + i_1 R + 0 R -$$

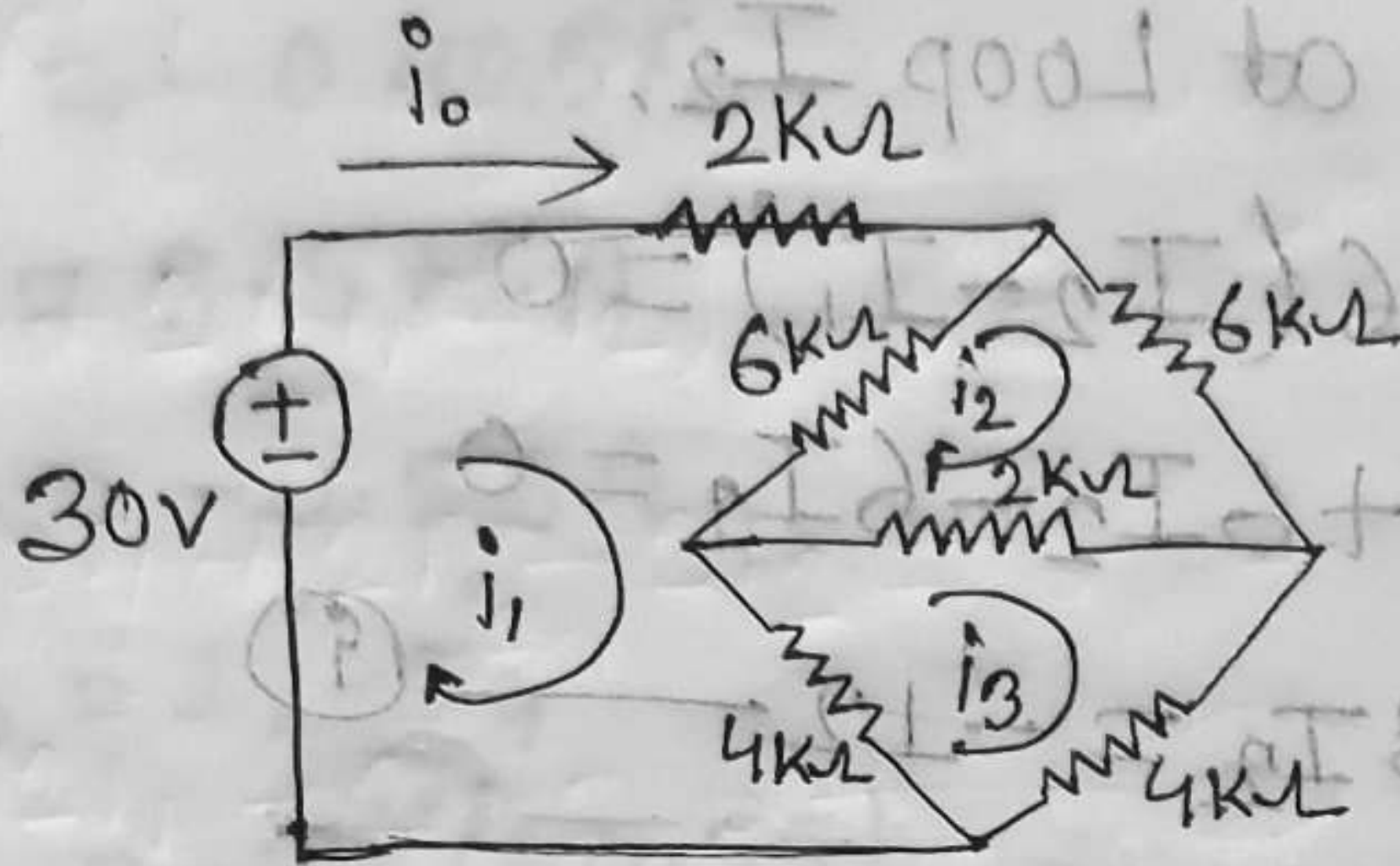


Apply KVL at Loop i_1

$$-30 + 2 \text{ (ii)}$$

$$0 = (i_1 - i_2) R + (i_1 - i_2) R + i_1 R$$

40.



$$i_0 = i_1$$

Apply KVL at Loop i_1 ,

$$-30 + 2i_1 + 6(i_1 - i_2) + 4(i_2 - i_3) = 0$$

$$\Rightarrow -30 + 2i_1 + 6i_1 - 6i_2 + 4i_2 - 4i_3 = 0$$

$$\Rightarrow 12i_1 - 6i_2 - 4i_3 = 30 \quad \text{--- (i)}$$

Apply KVL at Loop i_2 ,

$$6(i_2 - i_1) + 2(i_2 - i_3) + 6i_2 = 0$$

$$\Rightarrow 6i_2 - 6i_1 + 2i_2 - 2i_3 + 6i_2 = 0$$

$$\Rightarrow -6i_1 + 14i_2 - 2i_3 = 0 \quad \text{--- (ii)}$$

Apply KVL at Loop i_3 ,

$$4i_3 + 4(i_3 - i_1) + 2(i_3 - i_2) = 0$$

$$\Rightarrow 4i_3 + 4i_3 - 4i_1 + 2i_3 - 2i_2 = 0$$

$$\Rightarrow -4i_1 - 2i_2 + 10i_3 = 0 \quad \text{--- (iii)}$$

$$\therefore i_1 = 4.28 \text{ A} \quad \text{--- (ii)}$$

$$\therefore i_2 = 2.14 \text{ A}$$

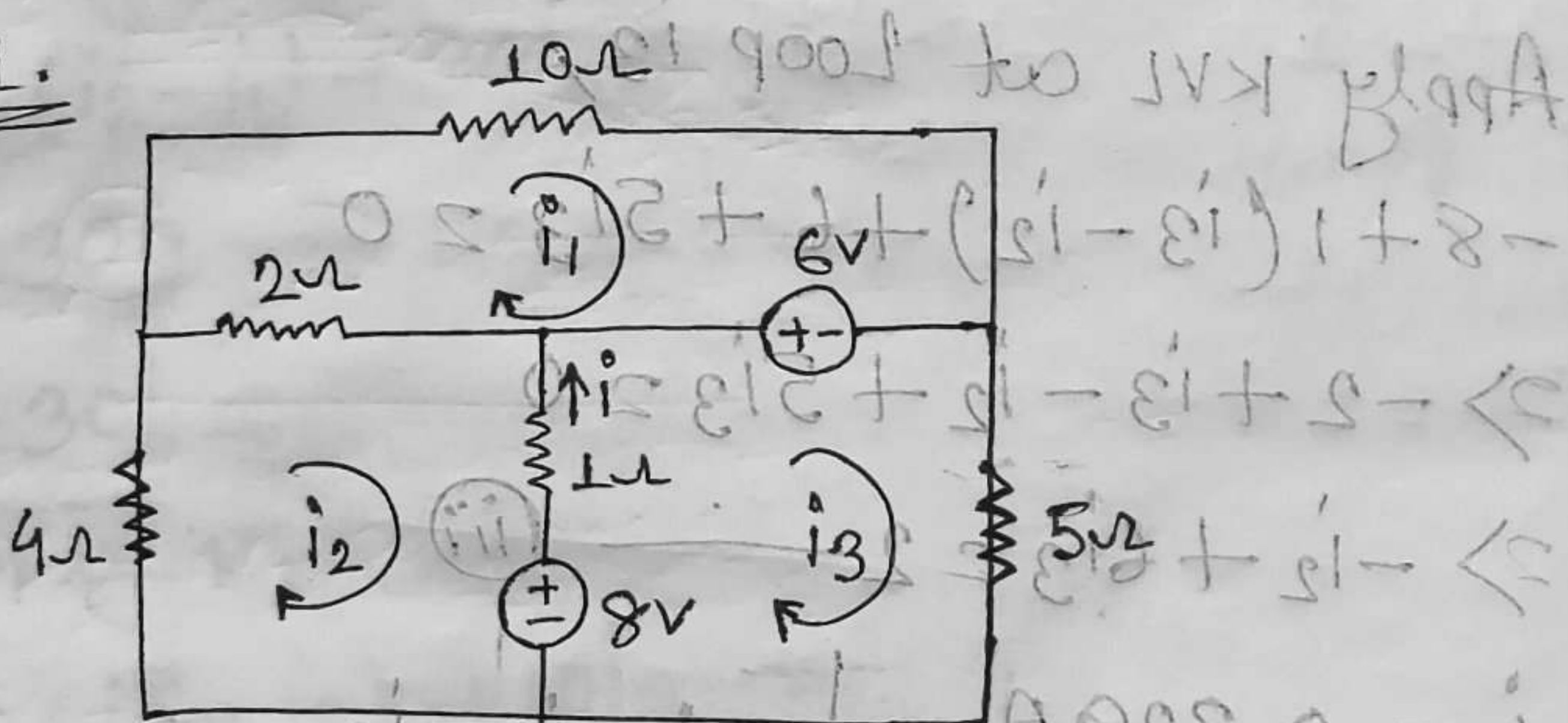
$$\therefore i_3 = 2.14 \text{ A}$$

$$\therefore i_o = i_1$$

$$= 4.28 \text{ A}$$

$$\text{--- (ii) } \underline{\underline{Am}} = i_1 - i_2 + i_3 =$$

41.



$$i = i_3 - i_2$$

$$A881.1 \leq$$

A

Apply KVL at Loop i_1 $\rightarrow$ $\epsilon i_1 + \epsilon i_1 \leftarrow$

$$10i_1 + 2(i_1 - i_2) - 6 = 0 \quad \epsilon i_1 - i_1 \leftarrow$$

$$\Rightarrow 10i_1 + 2i_1 - 2i_2 = 6 \quad \text{Ass. } i_1 = i_1$$

$$\Rightarrow 12i_1 - 2i_2 = 6 \quad \text{--- (i)}$$

Apply KVL at Loop i_2 $\text{Ass. } i_2 = i_2$

$$4i_2 + 2(i_2 - i_1) + 1(i_2 - i_3) + 8 = 0$$

$$\Rightarrow 4i_2 + 2i_2 - 2i_1 + i_2 - i_3 + 8 = 0$$

$$\Rightarrow -2i_1 + 7i_2 - i_3 = -8 \quad \text{--- (ii)}$$

Apply KVL at Loop i_3 $\text{Ass. } i_3 = i_3$

$$-8 + 1(i_3 - i_2) + 6 + 5i_3 = 0$$

$$\Rightarrow -2 + i_3 - i_2 + 5i_3 = 0$$

$$\Rightarrow -i_2 + 6i_3 = 2 \quad \text{--- (iii)}$$

$$i_1 = 0.329 \text{ A}$$

$$i_2 = -1.026 \text{ A}$$

$$i_3 = 0.162 \text{ A}$$

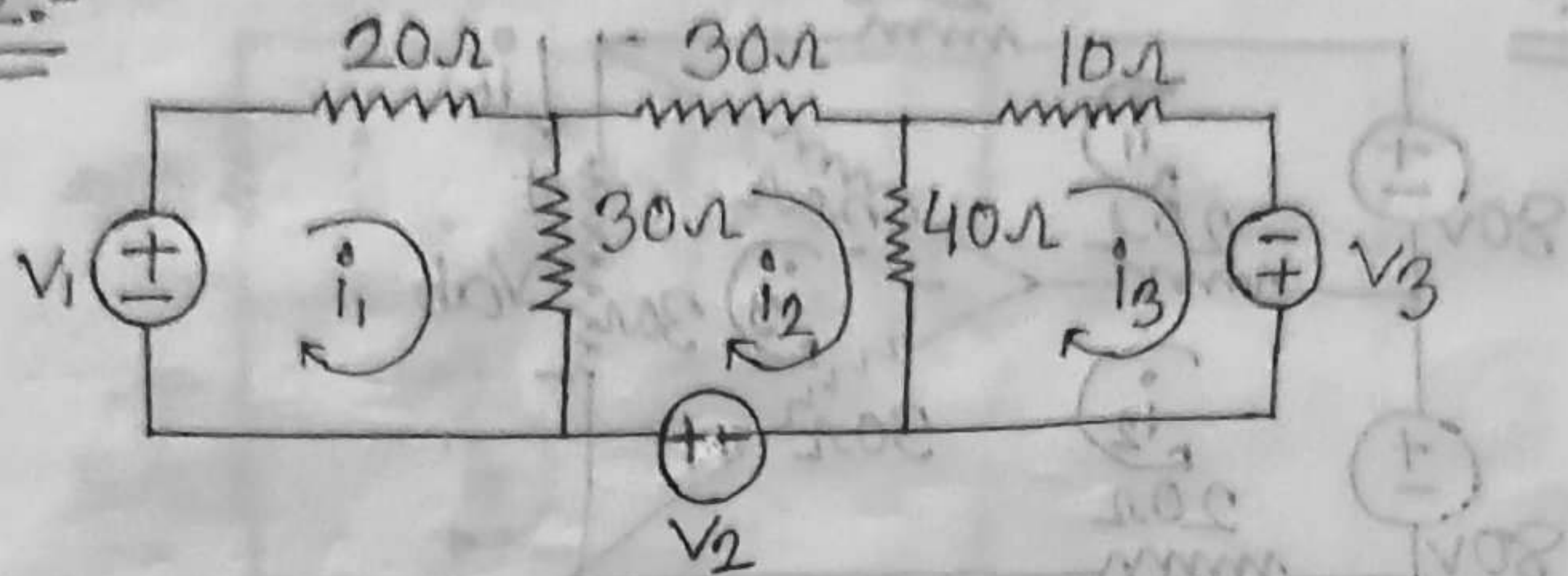
$$\therefore i_2 i_3 - i_2$$

$$= 0.162 - (-1.026)$$

$$= 1.188 \text{ A}$$

A

42:-



Apply KVL at Loop 1

$$-V_1 + 20i_1 + 30(i_1 - i_2) = 0$$

$$\Rightarrow -V_1 + 20i_1 + 30i_1 - 30i_2 = 0$$

$$\Rightarrow 50i_1 - 30i_2 + 0i_3 = V_1 \quad \text{--- (i)}$$

Apply KVL at Loop 2

$$30(i_2 - i_1) + 30i_2 + 40(i_2 - i_3) - V_2 = 0$$

$$\Rightarrow 30i_2 - 30i_1 + 30i_2 + 40i_2 - 40i_3 - V_2 = 0$$

$$\Rightarrow 30i_1 + 100i_2 - 40i_3 = V_2 \quad \text{--- (ii)}$$

Apply KVL at Loop 3

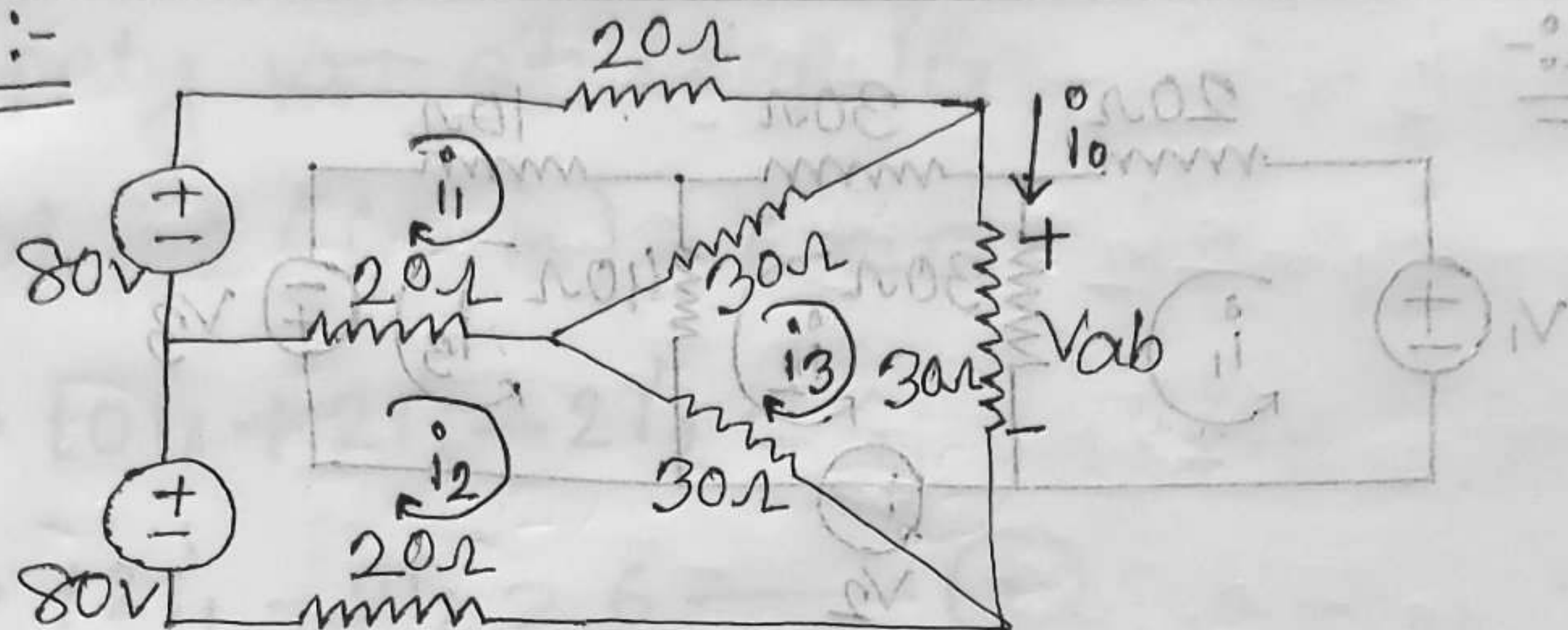
$$40(i_3 - i_2) + 10i_3 - V_3 = 0$$

$$\Rightarrow 40i_3 - 40i_2 + 10i_3 - V_3 = 0$$

$$\Rightarrow 0i_1 - 40i_2 + 50i_3 = V_3 \quad \text{--- (iii)}$$

A

43:-



Apply KVL at mesh 1,

$$-80 + 20i_1 + 30(i_1 - i_3) + 20(i_1 - i_2) = 0$$

$$\Rightarrow 70i_1 - 20i_2 - 30i_3 = 80 \quad \text{--- (i)}$$

Apply KVL at mesh 2,

$$-80 + 20(i_2 - i_1) + 20i_2 + 30(i_2 - i_3) = 0$$

$$\Rightarrow -20i_1 + 70i_2 - 30i_3 = 80 \quad \text{--- (ii)}$$

Apply KVL at mesh 3,

$$30i_3 + 30(i_3 - i_2) + 30(i_3 - i_1) = 0$$

$$30i_3 + 30(i_3 - i_2) + 30(i_3 - i_1) = 0 \quad \text{--- (iii)}$$

$$\Rightarrow -30i_1 - 30i_2 + 90i_3 = 0$$

$$i_1 = 2.67A$$

$$i_2 = 2.67A$$

$$i_3 = 1.78A$$

$$i_0 = i_3$$

$$= 1.78A$$

$$V_{ab} = i_0 \times R$$

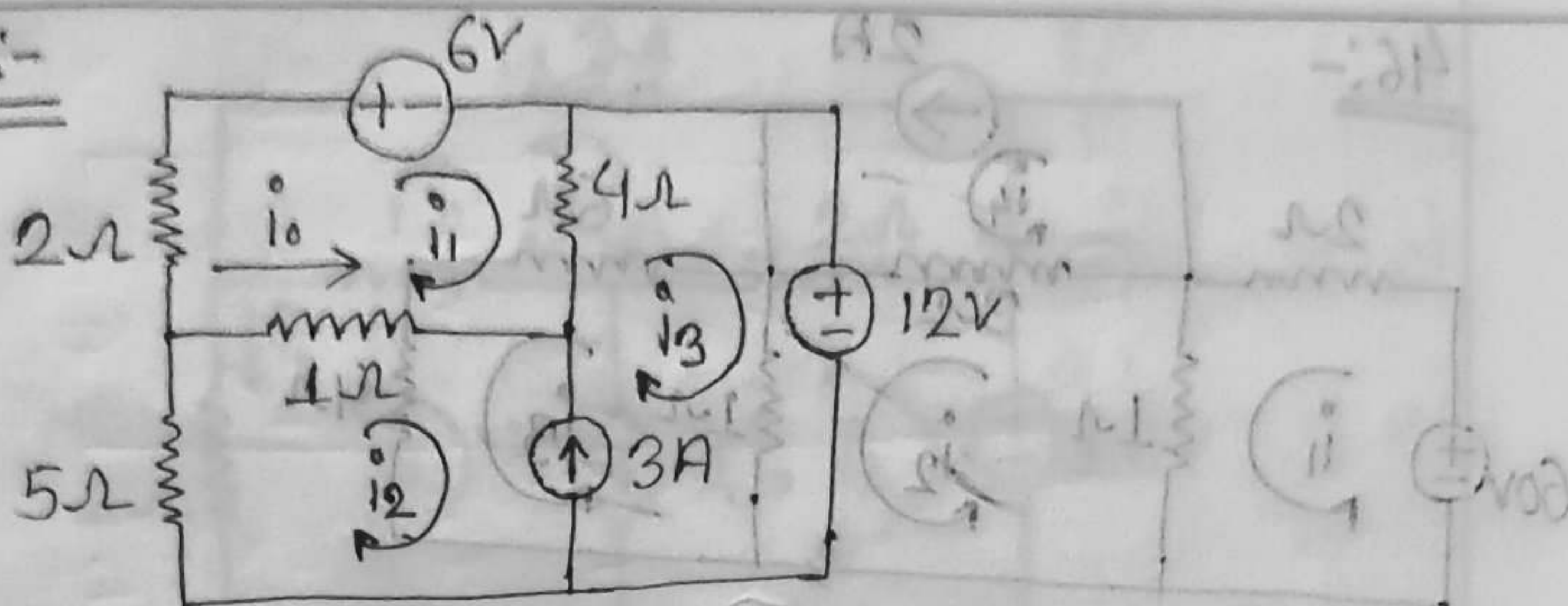
$$R = 30\Omega$$

$$= 1.78 \times 30$$

$$= 53.4V$$

Ans

44:-



$$i_0 = i_2 - i_1$$

$$i_3 - i_2 = 3 \quad \text{--- (i)}$$

Apply KVL at super mesh,

$$5i_2 + 1(i_2 - i_1) + 4(i_3 - i_1) + 12 = 0$$

$$\Rightarrow -5i_1 + 6i_2 + 4i_3 = -12 \quad \text{--- (ii)}$$

Apply KVL at mesh 1,

$$2i_1 + 6 + 4(i_1 - i_3) + 1(i_1 - i_2) = 0$$

$$\Rightarrow 7i_1 - i_2 - 4i_3 = -6 \quad \text{--- (iii)}$$

$$i_1 = -1.33A$$

$$i_2 = -3.067A$$

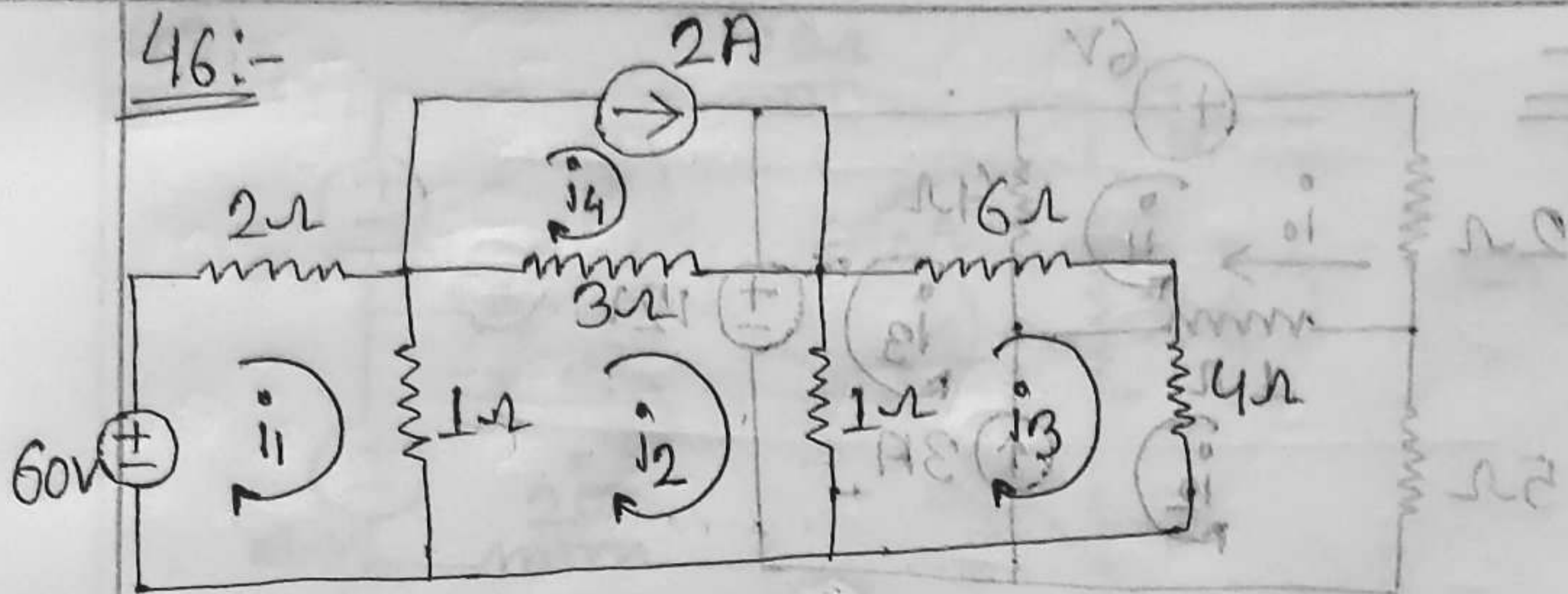
$$i_3 = -0.067A$$

$$\therefore i_0 = i_2 - i_1$$

$$= -3.067 - (-1.33) = -1.737A$$

Ans

46:-



$$i_4 = 2A$$

Apply KVL at Loop 1,

$$-60 + 2i_1 + 1(i_1 - i_2) = 0$$

$$\Rightarrow 3i_1 - 2i_2 = 60 \quad \text{--- (i)}$$

Apply KVL at Loop 2,

$$1(i_2 - i_1) + 3(i_2 - i_4) + 1(i_2 - i_3) = 0$$

$$\Rightarrow i_1 + 5i_2 - i_3 = 6 \quad \text{--- (ii)}$$

Apply KVL at Loop 3,

$$4i_3 + 1(i_3 - i_2) + 6i_3 = 0$$

$$\Rightarrow -i_2 + 11i_3 = 0 \quad \text{--- (iii)}$$

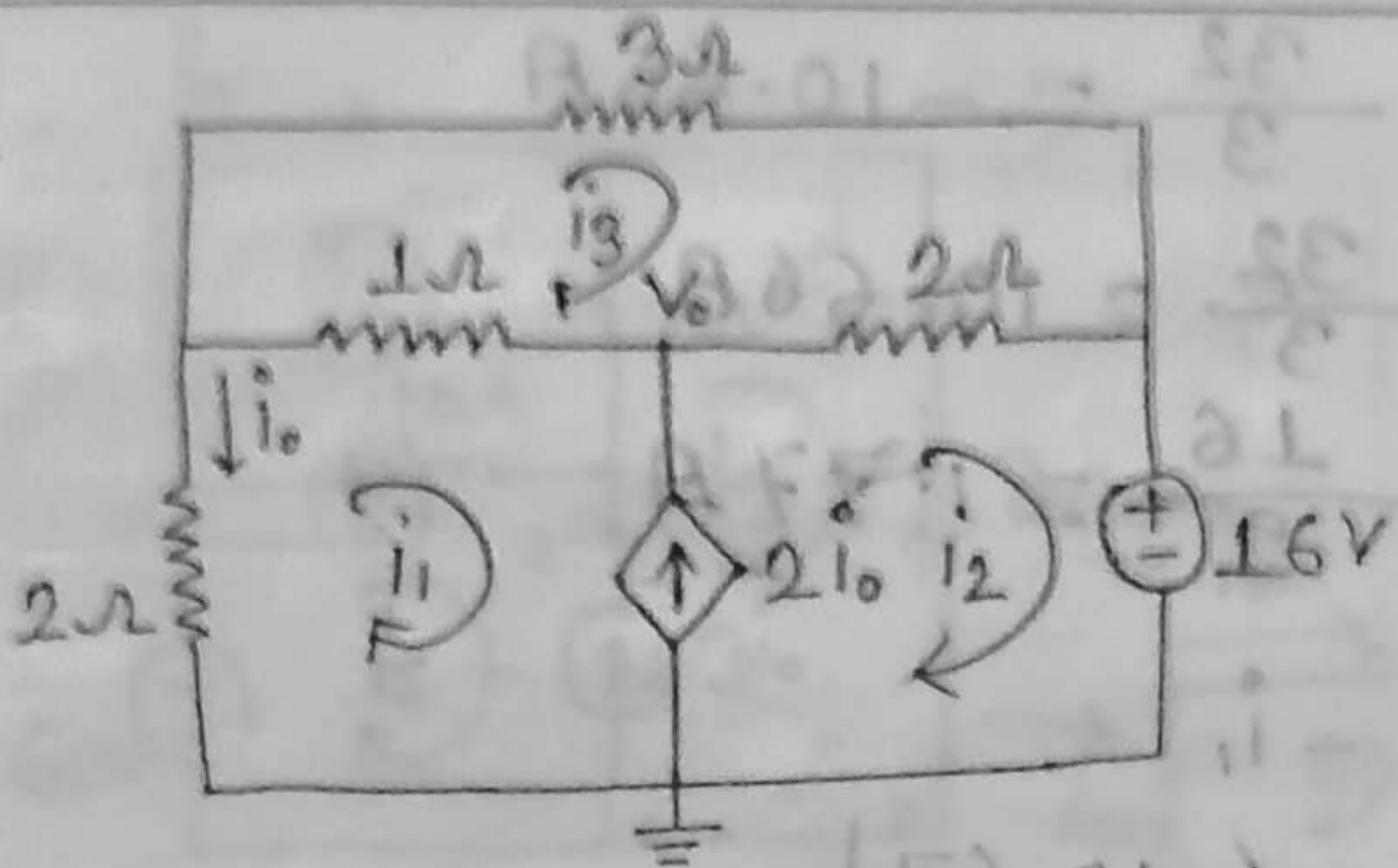
$$i_1 = 18.7A$$

$$i_2 = -2.93A$$

$$i_3 = -1.96A$$

Ans

49:-



$$i_o = -i_1$$

$$i_2 - i_1 = 2i_o$$

$$\Rightarrow i_2 - i_1 = -2i_1$$

$$\Rightarrow i_1 + i_2 = 0 \quad \text{--- (i)}$$

Apply KVL at super mesh,

$$2i_1 + 1i_1 - i_3 + 2i_2 - 2i_3 + 16 = 0$$

$$\Rightarrow 3i_1 + 2i_2 - 3i_3 = -16 \quad \text{--- (ii)}$$

Apply KVL at Loop i_3 ,

$$3i_3 + 2i_3 - 2i_2 + i_3 - i_1 = 0$$

$$\Rightarrow -i_1 - 2i_2 + 6i_3 = 0 \quad \text{--- (iii)}$$

$$i_1 = -\frac{32}{3} = -10.66 \text{ A}$$

$$i_2 = \frac{32}{3} = 10.66 \text{ A}$$

$$i_3 = \frac{16}{9} = 1.77 \text{ A}$$

$$\therefore i_0 = -i_1$$

$$= -(-10.67)$$

$$= 10.67 \text{ A}$$

$$I = i_2 - i_3$$

$$= 10.66 - 1.77$$

$$= 8.9 \text{ A}$$

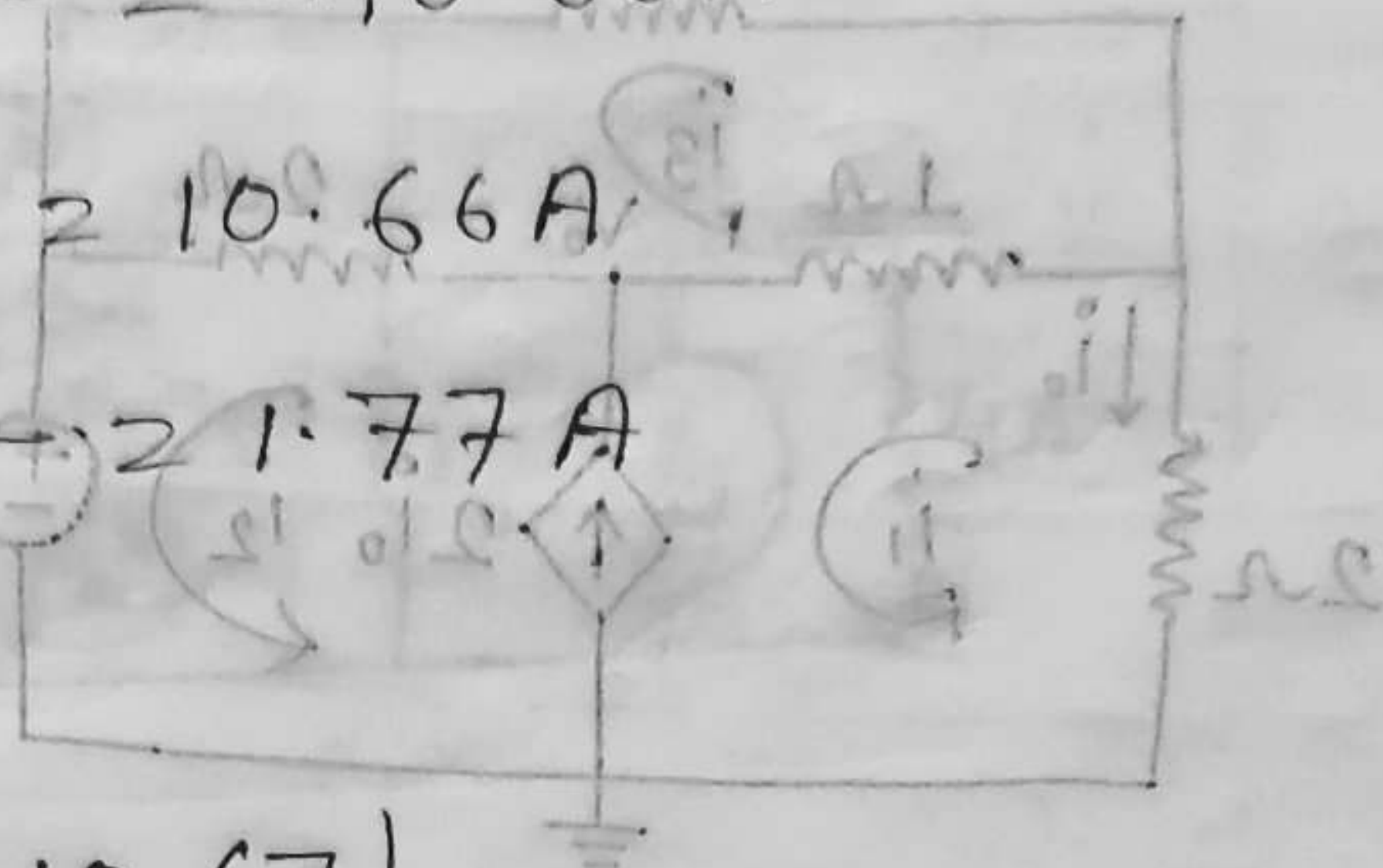
$$I = \frac{V_0 - 16}{2}$$

$$\Rightarrow 17.8 = V_0 - 16$$

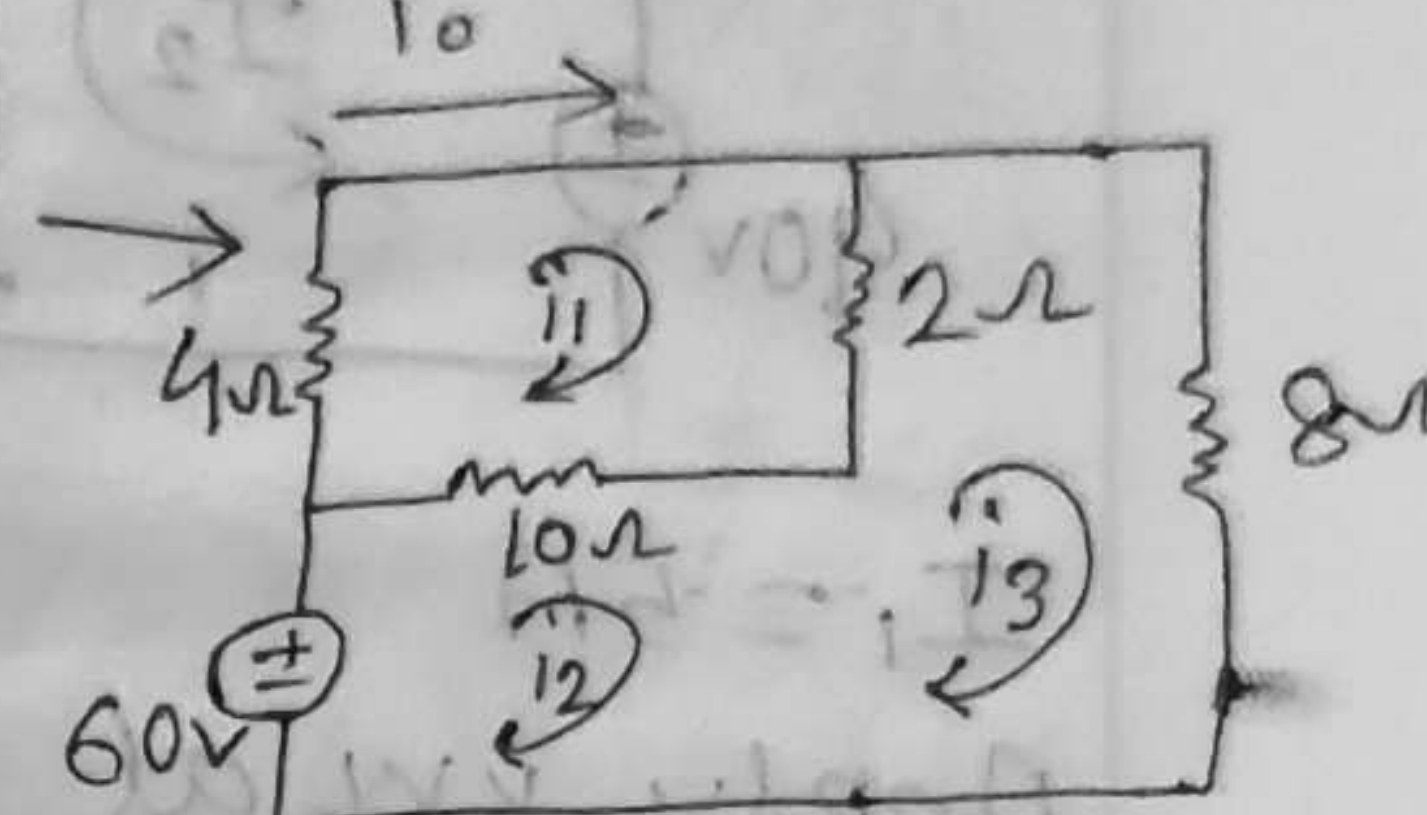
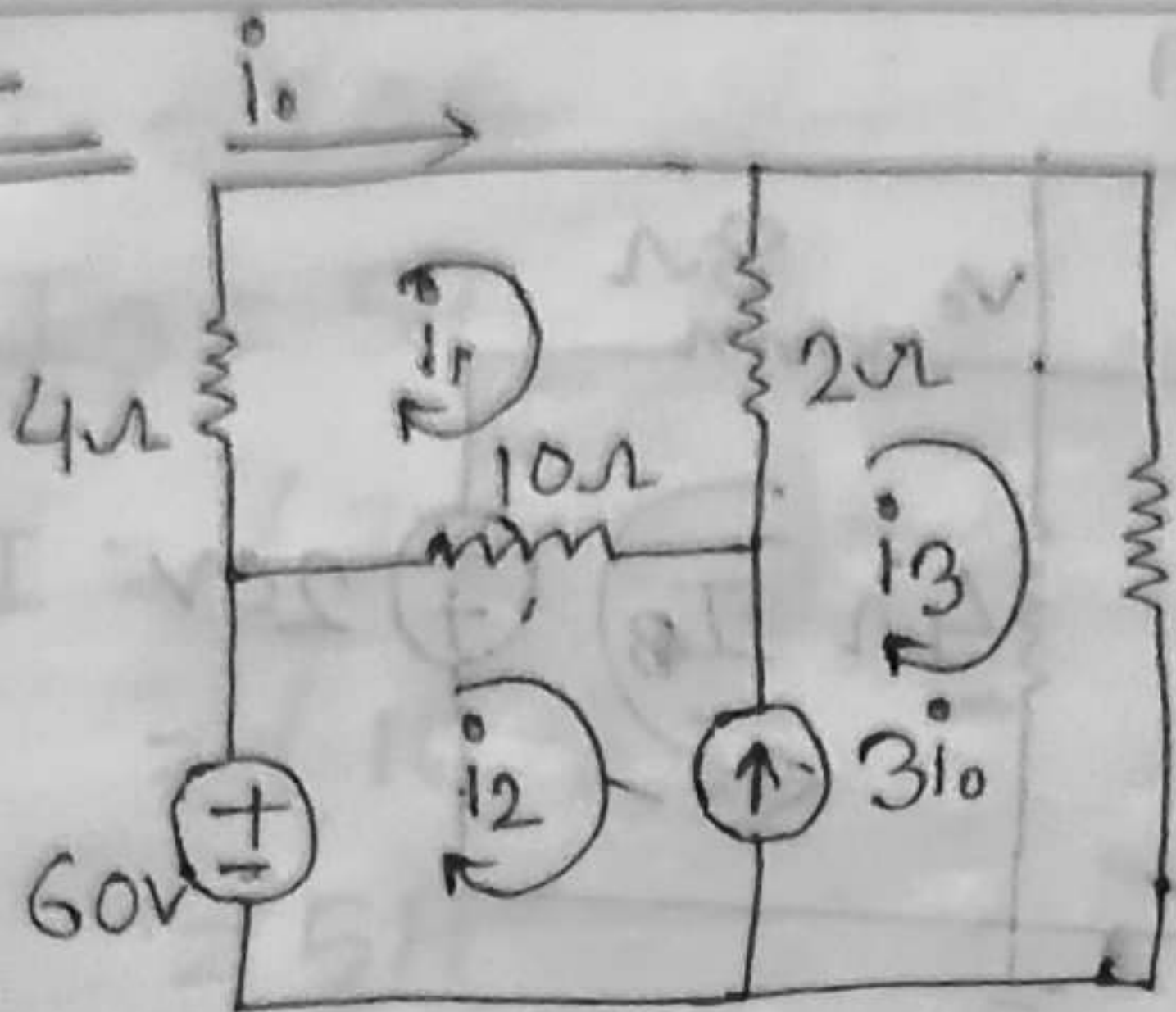
$$\Rightarrow V_0 = 17.8 + 16$$

$$\therefore V_0 = 33.8 \text{ V}$$

Ans



50:-



$$i_o = i_1$$

$$i_3 - i_2 = 3i_o$$

$$-3i_1 + i_3 - i_2 = 0$$

$$\Rightarrow -3i_1 - i_2 + i_3 = 0 \quad (i)$$

Apply KVL at super mesh,

$$-60 + 10(i_2 - i_1) + 8i_3 + 2(i_3 - i_1) = 0$$

$$\Rightarrow -12i_1 + 10i_2 + 10i_3 = 60 \quad (ii)$$

Apply KVL at Loop 1,

$$10(i_1 - i_2) + 4i_1 + 2(i_1 - i_3) = 0$$

$$\Rightarrow 16i_1 - 10i_2 - 2i_3 = 0 \quad (iii)$$

$$i_1 = 1.73A$$

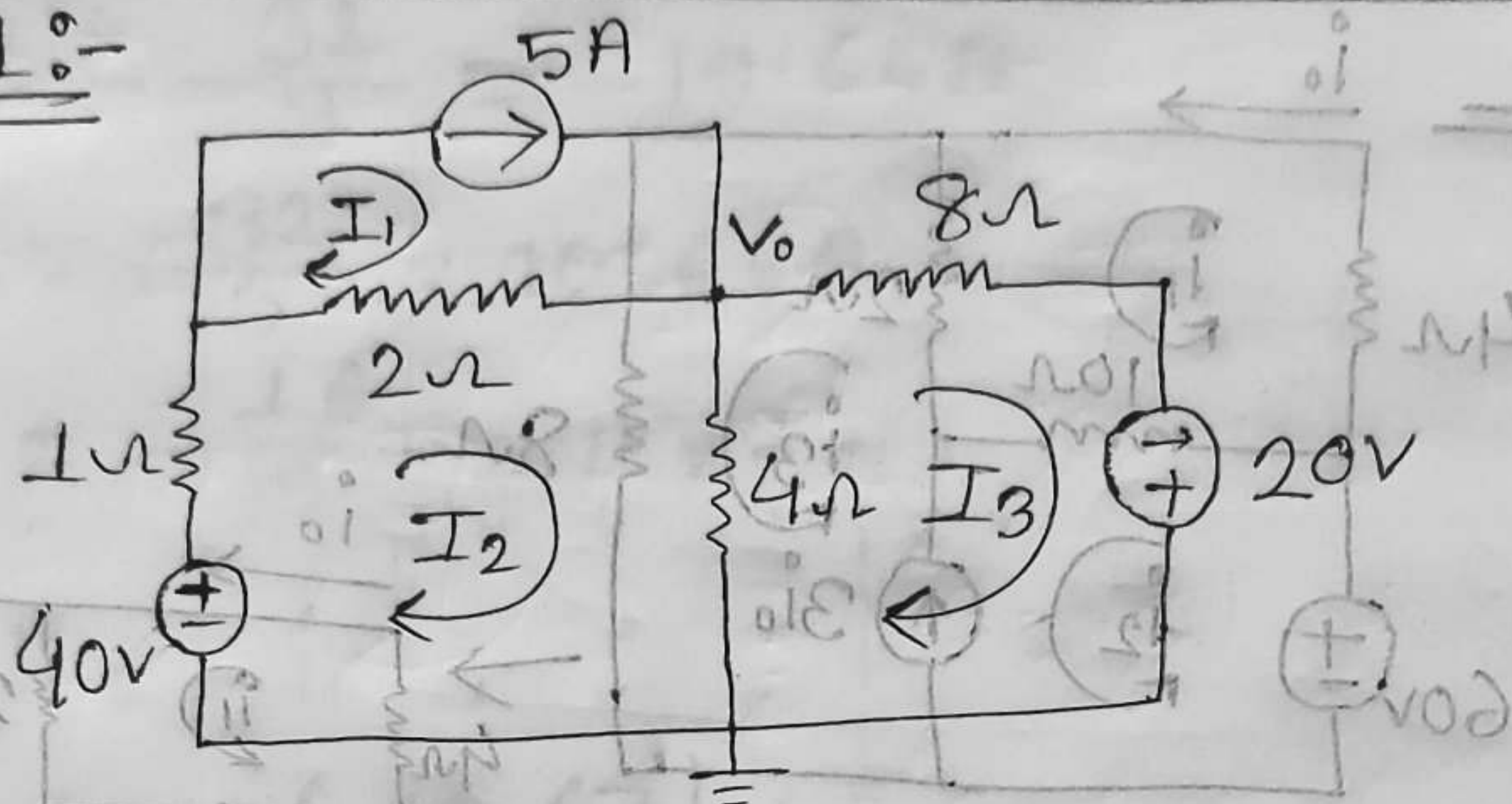
$$i_2 = 1.44A$$

$$i_3 = 6.63A$$

$$i_o = i_1 = 1.73A$$

Ans

51:-



$$I_1 = 5A$$

Apply KVL at Loop I_2 ,

$$-40 + 1I_2 + 2(I_2 - I_1) + 4(I_2 - I_3) = 0$$

$$\Rightarrow -40 + I_2 + 2I_2 - 2I_1 + 4I_2 - 4I_3 = 0$$

$$\Rightarrow -40 + 7I_2 - 2 \times 5 - 4I_3 = 0$$

$$\Rightarrow 7I_2 - 4I_3 = 50 \quad \text{--- (i)}$$

Apply KVL at Loop I_3 ,

$$(I_3 - I_2) + 8I_3 - 20 = 0$$

$$\Rightarrow 4I_3 - 4I_2 + 8I_3 - 20 = 0$$

$$\Rightarrow -4I_2 + 12I_3 = 20 \quad \text{--- (ii)}$$

$$\therefore I_2 = 10A$$

$$\therefore I_3 = 5A$$

$$\therefore I = (I_2 - I_3) \\ = (10 - 5) \\ = 5A$$

$$I = \frac{V_0 - 0}{4}$$

$$\Rightarrow 5 \times 4 = V_0$$

$$\therefore V_0 = 20V$$

Ans

$$0 = i_1 \cdot 8 + (i_1 - i_2) \cdot 4 + (i_1 - i_2) \cdot 2$$

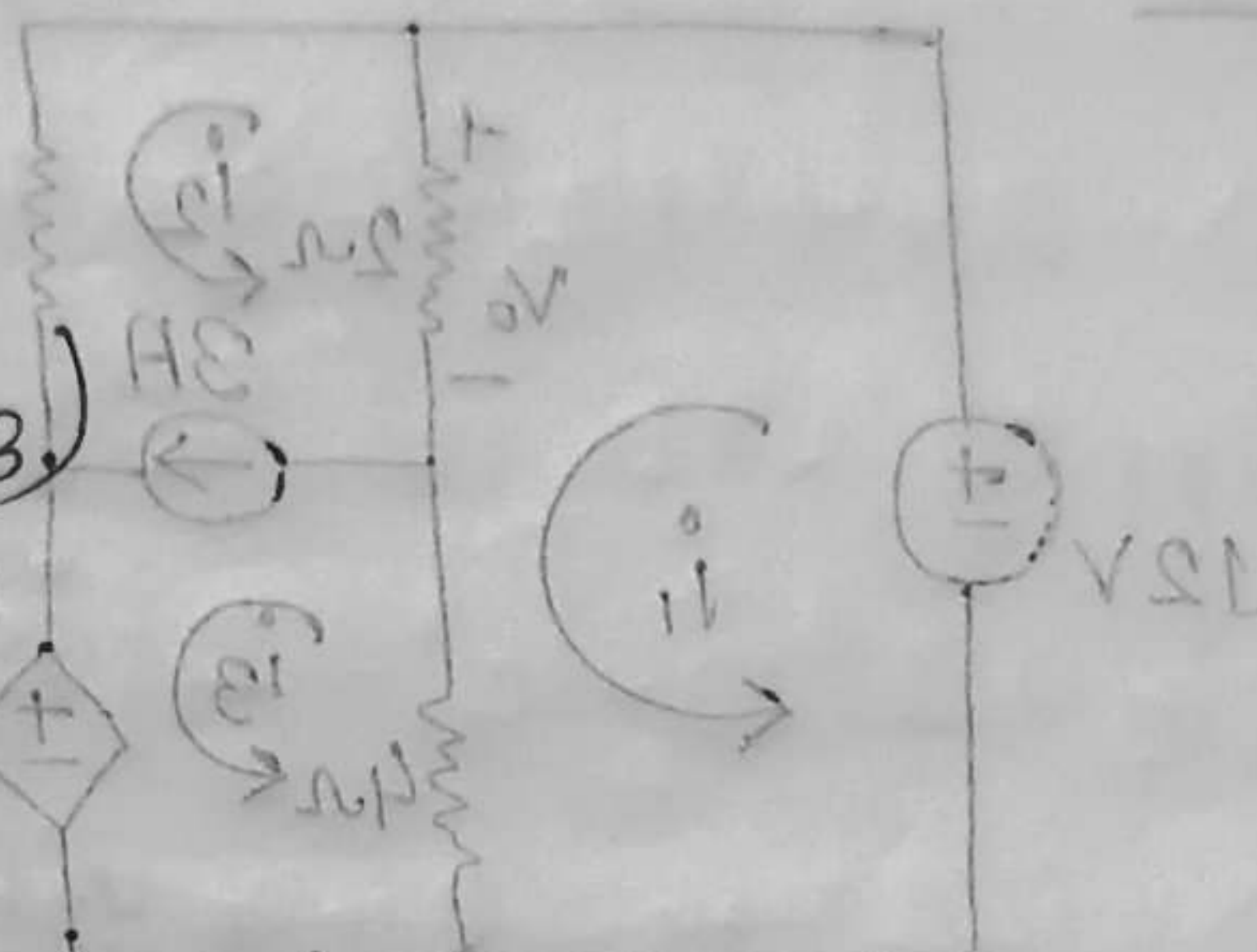
$$0 = 8i_1 + (i_1 - i_2) \cdot 4 + (i_1 - i_2) \cdot 2$$

$$\textcircled{ii} \quad 0 = i_1 \cdot 2 + i_1 \cdot 2 + i_2 \cdot 2 -$$

Apply KVL to loop 2

$$0 = i_1 - (i_1 - i_2) \cdot 4 + (i_1 - i_2) \cdot 2$$

$$\textcircled{iii} \quad i_1 = i_1 \cdot 2 - i_2 \cdot 2 - i_2 \cdot 2$$

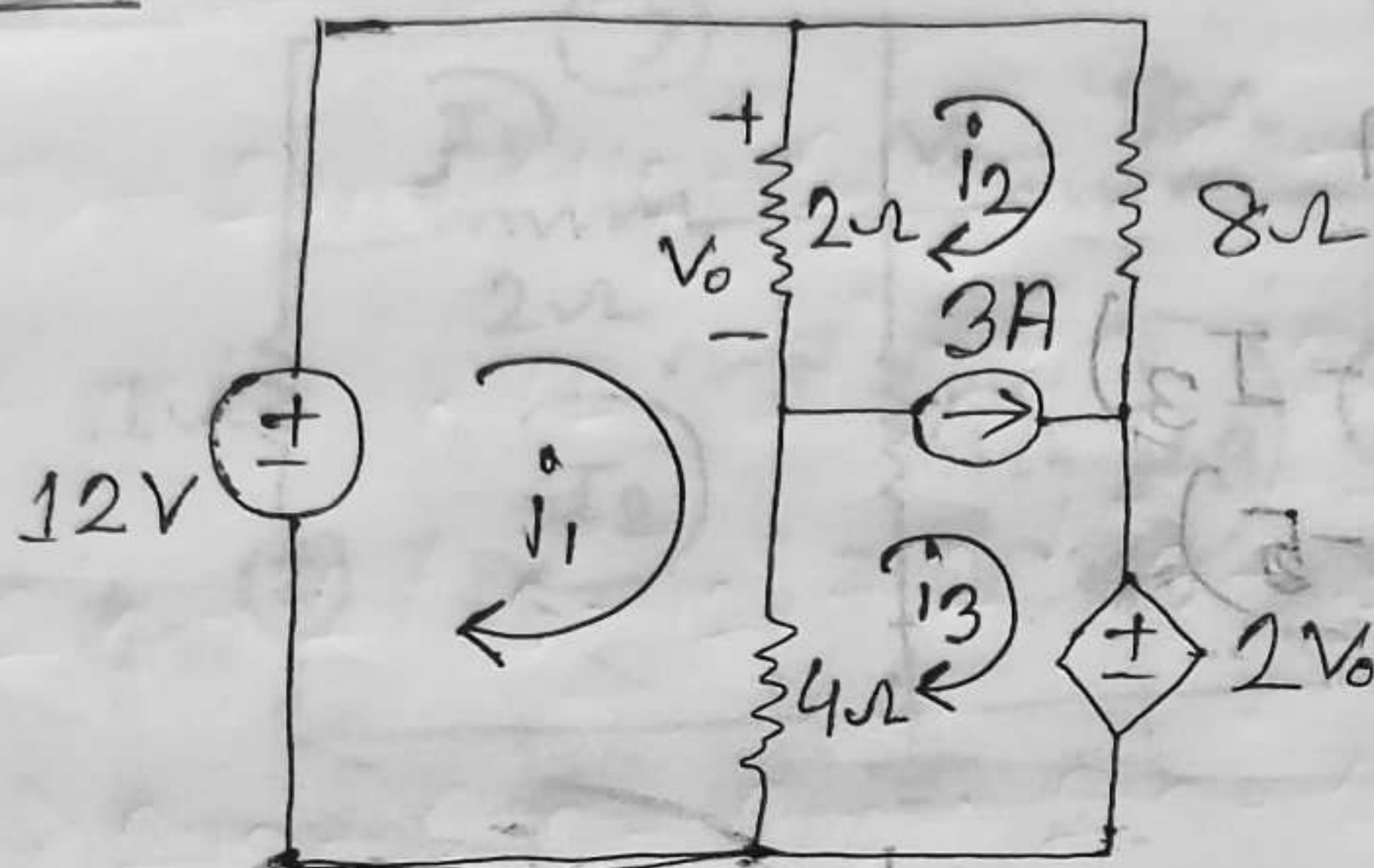


$$V = V_0 - 0$$

$$R = 4\Omega \Rightarrow V$$

$$\textcircled{i} \quad i_3 = i_1 - i_2$$

52:-



$$V_o = 2(i_1 - i_2)$$

$$i_3 - i_2 = 3 \quad \text{--- (i)}$$

Apply KVL at super mesh

$$2(i_2 - i_1) + 4(i_3 - i_1) + 2V_o + 8i_2 = 0$$

$$\Rightarrow 2i_2 - 2i_1 + 4i_3 - 4i_1 + 2 \times 2(i_1 - i_2) + 8i_2 = 0$$

$$\Rightarrow -2i_1 + 6i_2 + 2i_3 = 0 \quad \text{--- (ii)}$$

Apply KVL at Loop 1,

$$2(i_1 - i_2) + 4(i_1 - i_3) - 12 = 0$$

$$\Rightarrow 6i_1 - 2i_2 - 4i_3 = 12 \quad \text{--- (iii)}$$

$$i_1 = 3.5A$$

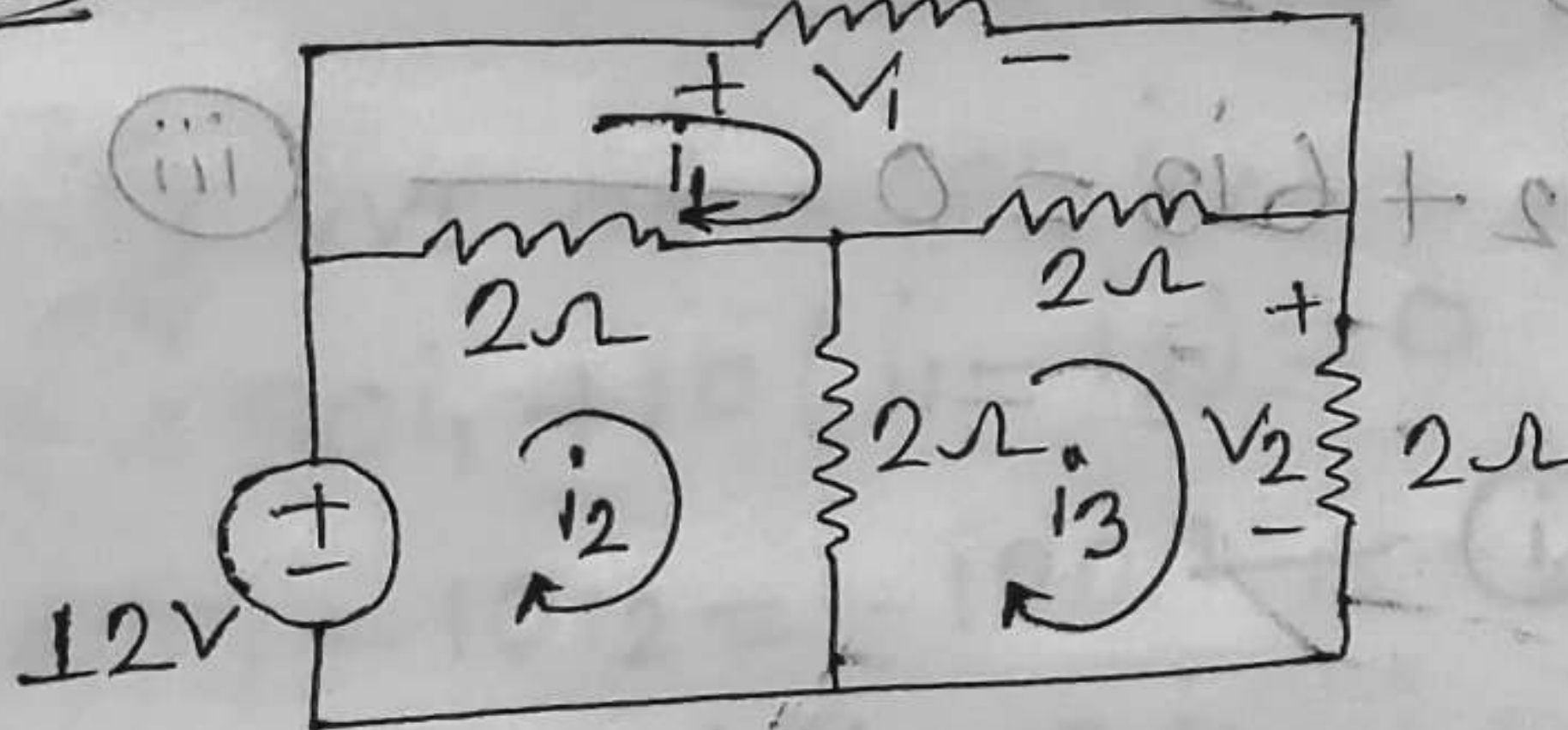
$$i_2 = -0.5A$$

$$i_3 = 2.5A$$

Ans

$$0 = (i_1 - i_2) \cdot 2 + (i_1 - i_2) \cdot 2 + i_1 \cdot 2$$

56:-



Apply KVL at Loop i_1

$$2i_1 + 2(i_1 - i_3) + 2(i_1 - i_2) = 0$$

$$\Rightarrow 2i_1 + 2i_1 - 2i_3 + 2i_1 - 2i_2 = 0$$

$$\Rightarrow 6i_1 - 2i_2 - 2i_3 = 0 \quad \text{--- (i)}$$

Apply KVL at Loop i_2

$$-12 + 2(i_2 - i_1) + 2(i_2 - i_3) = 0$$

Ans

$$\Rightarrow -12 + 2i_2 - 2i_1 + 2i_2 - 2i_3 = 0$$

$$\Rightarrow -2i_1 + 4i_2 - 2i_3 = 12 \quad \text{--- (ii)}$$

Apply KVL at Loop i_3 ,

$$2i_3 + 2(i_3 - i_2) + 2(i_3 - i_1) = 0$$

$$\Rightarrow 2i_3 + 2i_3 - 2i_2 + 2i_3 - 2i_1 = 0 \quad \text{--- (iii)}$$

$$\Rightarrow -2i_1 - 2i_2 + 6i_3 = 0$$

$$\therefore i_1 = 3A$$

$$\therefore i_2 = 6A$$

$$\therefore i_3 = 3A \quad \text{Here,}$$

$$V_1 = i_1 R_1$$

$$= 3 \times 2$$

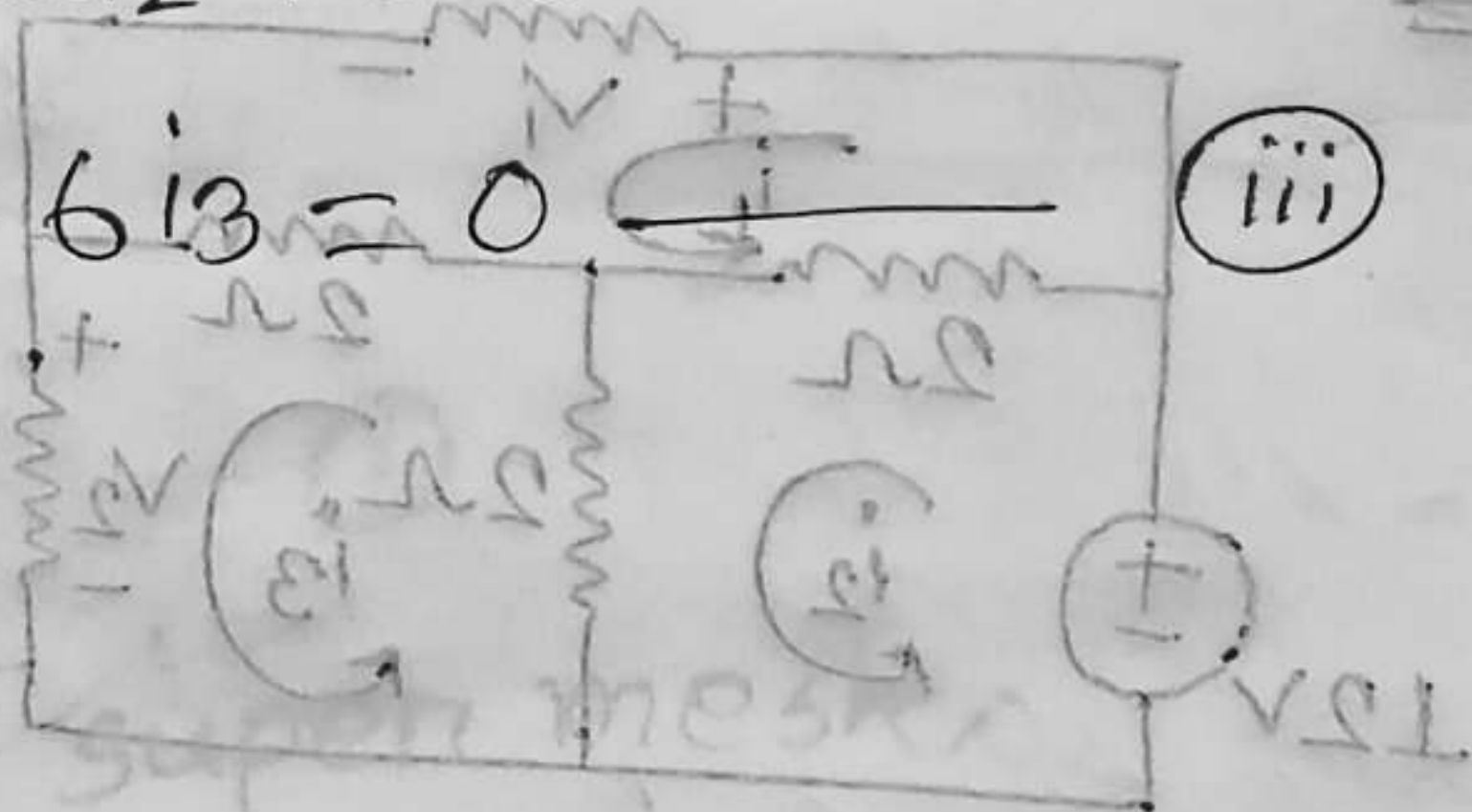
$$= 6V$$

$$V_2 = i_3 \times R_3$$

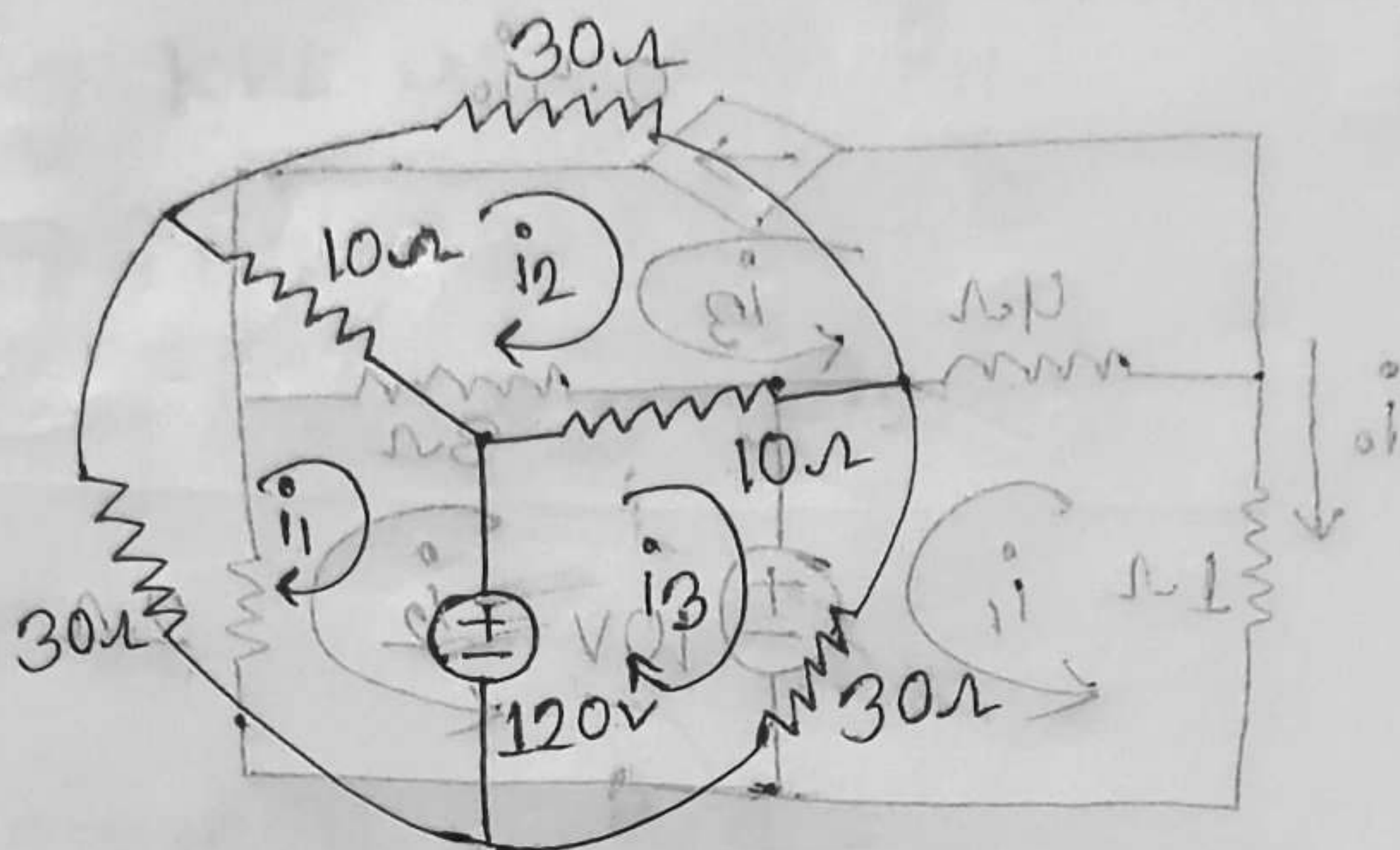
$$= 3 \times 2$$

$$= 6V.$$

Ans



58:-



Apply KVL at Loop i_1 ,

$$120 + 30i_1 + 10(i_1 - i_2) = 0$$

$$\Rightarrow 40i_1 - 10i_2 = -120 \quad \text{--- (i)}$$

Apply KVL at Loop i_2 ,

$$10(i_2 - i_3) + 10(i_2 - i_1) + 30i_2 = 0$$

$$\Rightarrow -10i_1 + 50i_2 - 10i_3 = 0 \quad \text{--- (ii)}$$

Apply KVL at Loop i_3 ,

$$-120 + 10(i_3 - i_2) + 30i_3 = 0$$

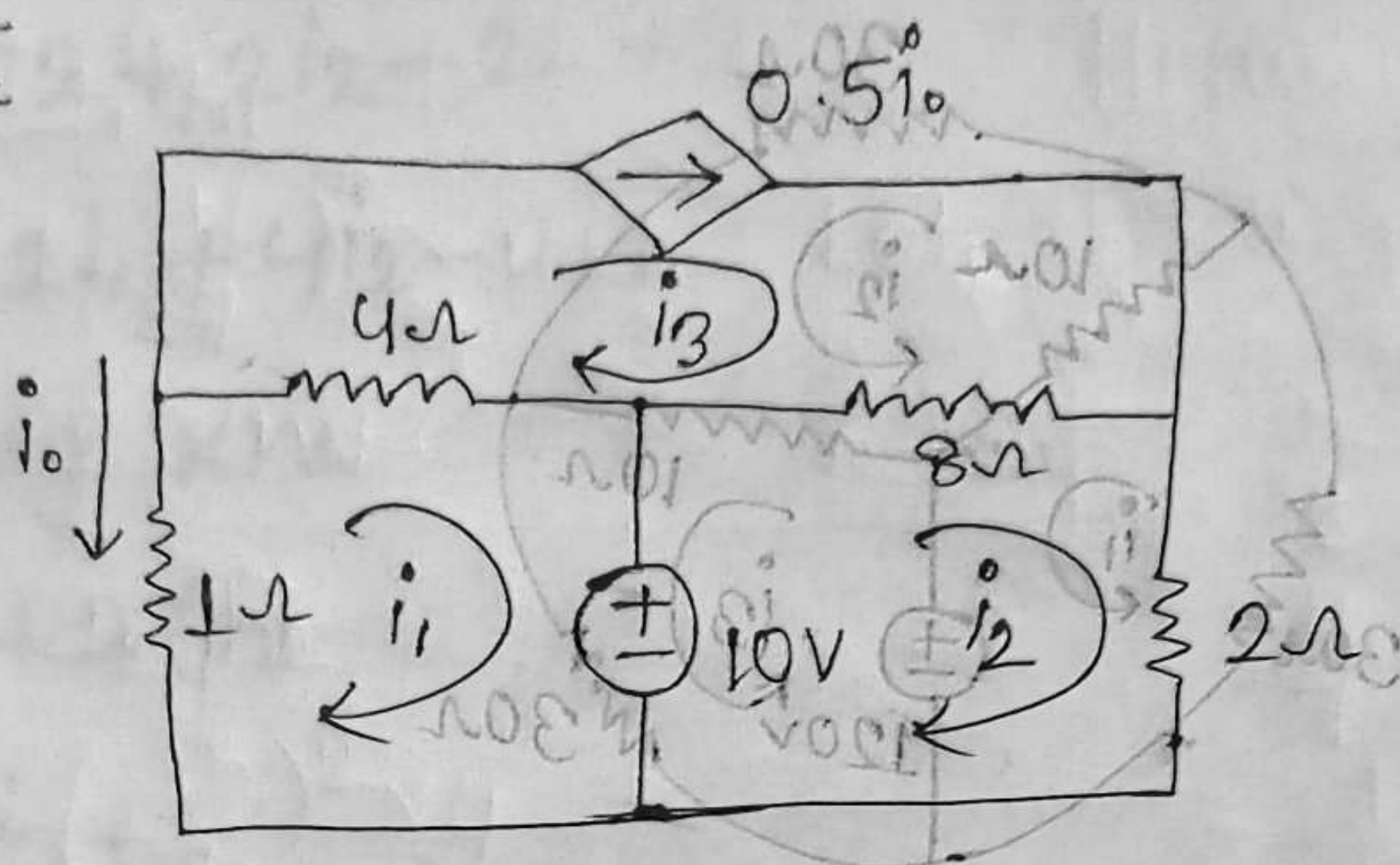
$$\Rightarrow -10i_2 + 40i_3 = 120 \quad \text{--- (iii)}$$

$$\therefore i_1 = -3A$$

$$\therefore i_2 = 0A$$

$$\therefore i_3 = 3A$$

60:-



$$i_3 = 0.5 i_o$$

$$i_o = -i_1$$

$$\Rightarrow i_3 = 0.5(-i_1) = -0.5i_1$$

Apply KVL at Loop 1:

$$i_1 + 4(i_1 - (-0.5i_1)) + 10 = 0$$

$$\Rightarrow i_1 + 4i_1 + 2i_1 + 10 = 0$$

$$\Rightarrow 7i_1 = -10$$

$$\therefore i_1 = -\frac{10}{7} \text{ A}$$

$$A_E = i_1$$

$$A_O = i_2$$

$$A_E = i_1$$

Apply KVL at Loop 2, $i_2 = 0$

$$-10 + 8(i_2 - i_3) + 2i_2 = 0 \quad \left(\frac{0.1}{7} -\right) =$$

$$\Rightarrow -10 + 8i_2 - 8i_3 + 2i_2 = 0 \quad \text{WFO.2} =$$

$$\Rightarrow 10i_2 - 8(-0.5i_1) = 10 \quad \text{WFO.2} =$$

$$\Rightarrow 10i_2 + 4i_1 = 10$$

$$4 \times \left(\frac{2.1}{7}\right) =$$

$$\Rightarrow 10i_2 + 4 \times \left(\frac{-10}{7}\right) = 10 \quad \text{WFO.81} =$$

$$\Rightarrow 10i_2 = 10 - 4 \times \left(\frac{-10}{7}\right)$$

$$8 \times \left(\frac{2.1}{7}\right) =$$

$$\Rightarrow 10i_2 = \frac{110}{7}$$

$$8 \times \left(\frac{2.1}{7}\right) =$$

$$\Rightarrow i_2 = \frac{11}{7}$$

$$WFO.2 =$$

$$\therefore i_2 = \frac{11}{7} \text{ A}$$

$$4 \times \left(\frac{11}{7}\right) =$$

$$i_0 = 8i_1 \quad \text{WFO.4} =$$

$$\therefore i_0 = \frac{10}{7}$$

mA

$$\therefore i_3 = -0.5i_1 = -0.5\left(\frac{-10}{7}\right) = \frac{5}{7} \text{ A.}$$

$$P_1 = I_1^2 R_1 \quad \text{is good to Here, } 299A$$

$$= \left(-\frac{10}{7}\right)^2 \times 1 \quad \text{Let } I_1 = i_1 = -\frac{10}{7} A$$

$$= 2.04 W \quad I_2 = i_3 - i_1$$

$$P_2 = I_2^2 R_2 \quad O.L = (i_2 - 0) \times 8 = \frac{5}{7} - \left(-\frac{10}{7}\right)$$

$$= \left(\frac{15}{7}\right)^2 \times 4 \quad O.L = i_1 = \frac{15}{7} A$$

$$= 18.36 W \quad O.L = \left(\frac{O.L}{F}\right) \quad I_3 = i_2 - i_3$$

$$P_3 = I_3^2 R_3 \quad \left(\frac{O.L}{F}\right) \times 12 - O.L = \frac{11}{7} - \frac{5}{7}$$

$$= \left(\frac{6}{7}\right)^2 \times 8 \quad O.L \therefore I_3 = \frac{6}{7} A$$

$$= 5.87 W \quad \text{Let } I_4 = i_2 = \frac{11}{7} A$$

$$P_4 = I_4^2 R_4$$

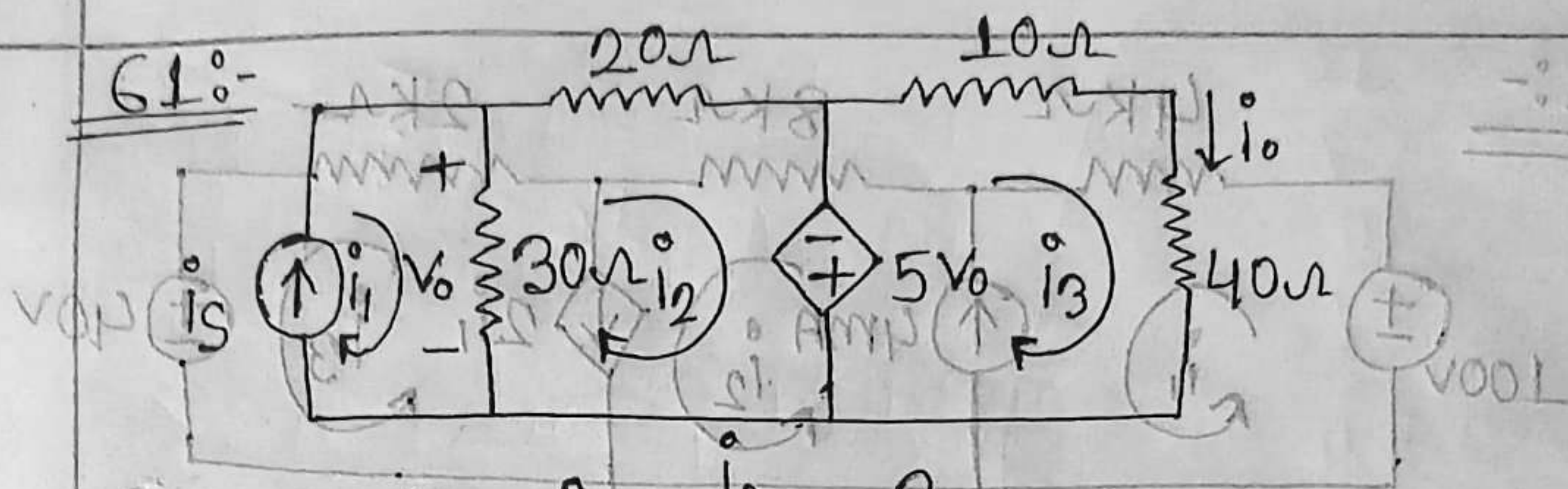
$$= \left(\frac{11}{7}\right)^2 \times 2$$

$$= 4.93 W$$

Ans

- A $R_1 = 1 \Omega$
- $R_2 = 4 \Omega$
- $R_3 = 8 \Omega$
- $\left(\frac{O.L}{F}\right) R_4 = 2 \Omega$

$$A \frac{O.L}{F} = \left(\frac{O.L}{F}\right) 2.0 - = i_2 - 0 = 8 A$$



Current gain $\frac{i_o}{i_s} = ?$

$$i_1 = i_s$$

$$i_3 = i_o$$

$$V_0 = IR$$

$$\therefore V_0 = (i_1 - i_2) 30$$

$$\text{and, } 5V_0 = 5 \times 30(i_1 - i_2) \\ = 150(i_1 - i_2)$$

Apply KVL at Loop 1

$$20i_2 - 5V_0 + 30(i_2 - i_1) = 0$$

$$\Rightarrow 20i_2 - 150i_1 + 150i_2 + 30i_2 - 30i_1 = 0$$

$$\Rightarrow -180i_1 + 200i_2 = 0$$

$$\Rightarrow 200i_2 = 180i_1$$

$$\Rightarrow i_2 = 0.9i_1$$

Apply KVL at Loop 3,

$$10i_3 + 40i_3 + 5V_0 = 0$$

$$\Rightarrow 50\dot{i}_3 + 150(\dot{i}_1 - \dot{i}_2) = 0$$

$$\Rightarrow 50\dot{i}_3 + 150(\dot{i}_1 - 0.9\dot{i}_1) = 0$$

$$\Rightarrow 50\dot{i}_3 + 150 \times 0.1\dot{i}_1 = 0$$

$$\Rightarrow 50\dot{i}_3 + 15\dot{i}_1 = 0$$

$$\Rightarrow \dot{i}_3 = -\frac{15}{50}\dot{i}_1$$

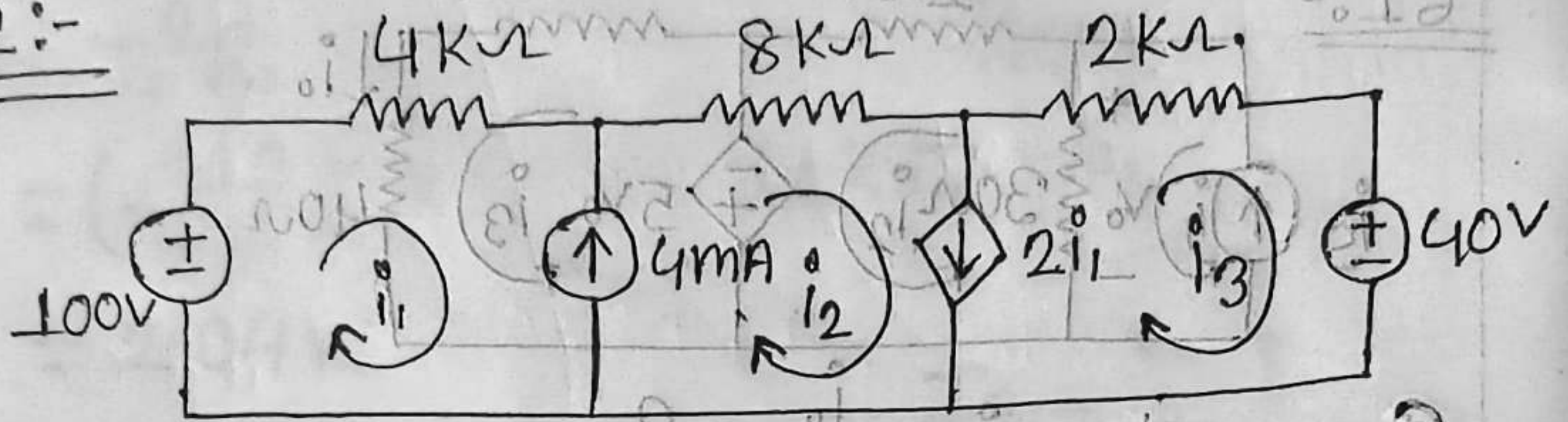
$$\Rightarrow \dot{i}_3 = -0.3\dot{i}_1$$

$$\therefore \dot{i}_0 = -0.3\dot{i}_s$$

$$\Rightarrow \frac{\dot{i}_0}{\dot{i}_s} = -0.3$$

Ans

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$$i_2 - i_1 = 4 \text{ mA} \quad \text{--- (i)}$$

$$i_2 - i_3 = 2i_1$$

$$\Rightarrow -2i_1 + i_2 - i_3 = 0 \quad \text{--- (ii)}$$

Apply KVL at super mesh,

$$-100 + 4i_1 + 8i_2 + 2i_3 + 40 = 0$$

$$\Rightarrow 4i_1 + 8i_2 + 2i_3 = 60 \quad \text{--- (iii)}$$

$$\therefore i_1 = 2 \text{ mA}$$

$$\therefore i_2 = 6 \text{ mA}$$

$$\therefore i_3 = 2 \text{ mA}$$

Ans